INTERNATIONAL MONETARY FUND

SOLOMON ISLANDS

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STAFF REPORT FOR THE 2017 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS[1]

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The external risk of debt distress in the Solomon Islands remains moderate, consistent with the 2016 assessment. Under the baseline scenario—adjusted for the average long-term effects of natural disasters on growth, the fiscal balance, and the current account balance— all external debt indicators remain below the relevant policy-dependent indicative thresholds. The baseline is subject to significant risks with sizable and protracted breaches of thresholds from a shock to financing terms. Adding domestic debt to the analysis does not alter the overall risk of debt distress, although the baseline rapidly approaches the indicative threshold under the public debt-to-GDP ratio towards the end of the projection period. That said, public debt is on an upward trajectory and could rise above the authorities' nominal public debt limit of 30 percent of GDP by 2028. A shock to real GDP growth would cause a significant breach of the public debt-to-GDP threshold.

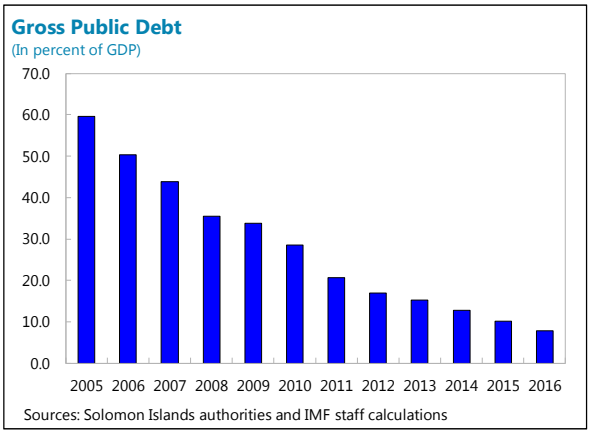
Since Solomon Islands is highly vulnerable to natural disasters, the analysis has been complemented with a stress test of a single severe natural disaster occurring in 2018—of similar scale to the largest shock in Solomon Islands' history. Debt sustainability deteriorates in the aftermath of the event and growth declines with public debt to GDP ratio rising close to the threshold.

Solomon Islands fiscal performance has deteriorated recently, adding to prospective vulnerability. Fiscal buffers have been eroded and domestic arrears are accumulating. Both revenue raising measures and expenditure control measures are required to place the fiscal position on a firmer footing and to enable Solomon Islands to respond to shocks.

[1] The policies and institutions of Solomon Islands, as measured by the CPIA, averaged 2.96 over the past three years. Since its average CPIA index has been below 3.25 for three years in a row, the country's policy performance remains classified as weak, according to the "Staff Guidance Note on the Application of the Joint Fund-Bank Debt Sustainability Framework for Low-income Countries" (www.imf.org/external/np/pp/eng/2010/012210.pdf). Therefore, the relevant indicative thresholds for this category are: 30 percent for the PV of debt-to-GDP ratio, 100 percent for the PV of debt-to-exports ratio, 200 percent for the PV of debt-to-revenue ratio, 15 percent for the debt service-to-exports ratio, and 18 percent for the debt service-to-revenue ratio. These thresholds are applicable to public and publicly guaranteed external debt.

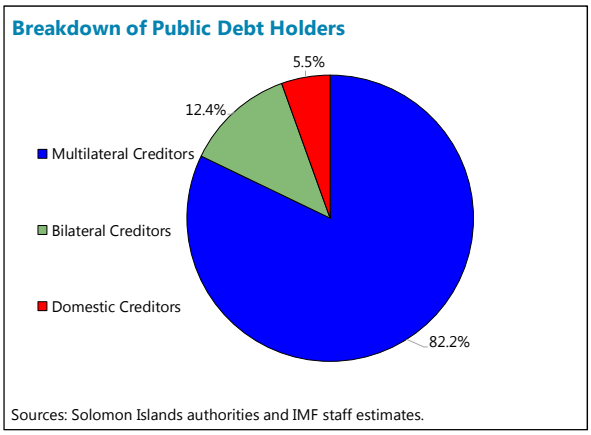
BACKGROUND

1. Good progress has been made over the past decade in building fiscal buffers and reducing public debt from 50.3 percent of GDP in 2006 to just 7.9 percent of GDP at end-2016. In 2005, the Honiara Club Agreement restructured around 10 percent of the stock of external public debt and a moratorium was placed on debt servicing and new external borrowing. Following further improvement in the debt position, the agreement was reviewed and a new debt management framework was introduced in 2012 which enabled the resumption of concessional external borrowing. In September 2016, the debt management framework was revised and strengthened—with new guidelines on direct borrowing, on-lending, and guarantees. The debt management strategy stipulates a limit for the public debt-to-GDP ratio and debt service to domestically sourced revenue ratio set at 30 percent and 10 percent respectively. The guidelines also require the government to aim for its risk of debt distress to be no more than moderate, as defined by the Debt Sustainability Analysis (DSA).



2. External Public and Publicly Guaranteed (PPG) debt stood at US$89 million (7.5 percent of GDP) in 2016 while private sector external debt amounted to just 0.8 percent of GDP. The private sector in the Solomon Islands has very limited capacity to borrow externally.

3. Multilateral and bilateral official creditors account for the bulk of all loans. The International Development Association (IDA) and the Asian Development Bank (ADB) account for 37.0 percent and 36.9 percent of total external PPG debt respectively, while the share of bilateral creditors—Taiwan Province of China and EU— is 13.1 percent of the total. Australia and New Zealand are important donors for the country, together providing over two-thirds of total Official Development Assistance to Solomon Islands.



Stock of external and domestic debt at end-2016

	In million of US dollars	As a share of total debt	In percent of GDP
Total external debt	98.7	100%	8.2%
of which: Public and Publicly Guaranteed (PPG)	89.3	91%	7.5%
of which: Private	9.4	9%	0.8%
	In million of SI dollars	As a share of total debt	In percent of GDP
Total public debt	774.8	100%	7.9%
of which: external	732.2	95%	7.5%
of which: domestic	42.6	5%	0.4%
Contingent Liabilities from SOEs	121.1	-	1.2%

Source: Solomon Islands authorities and IMF staff estimates.

4. Public domestic debt at end-2016 stood at SI$42.6 million (0.4 percent of GDP), most of which was in Treasury bills. In December 2015, the entire stock of government bonds, amounting to SI$99.5 million, was repaid ahead of schedule. The implicit contingent liabilities from state-owned enterprises were around SI$121 million (1.2 percent of GDP) at end-2016.

5. However, fiscal performance has deteriorated recently and borrowing is on an upward trajectory. The government tripled its annual borrowing limit from SI$300 million in the 2016 budget to SI$900 million in the 2017 budget; SI$600 million was set aside for the Tina River hydropower project, while the remaining SI$300 million has been earmarked for other purposes, including the undersea cable project. In March 2017, the government issued domestic development bonds, worth SI$150 million, which were purchased by the country's national pension fund, the Solomon Islands National Provident Fund. This increased the level of public debt to about 9.4 percent of GDP. The government also intends to issue additional bonds amounting to SI$150 million by end-2018. Disbursements for the Tina River hydropower development projects are expected to begin in 2018 (Box 1), for which in early 2017, the government also provided guarantees, worth US$15.4 million (about SI$122 million).

Box 1. Tina River Hydropower Development Project (TRHDP)

Solomon Islands' first large-scale energy project, aims to reduce electricity costs—one of the highest in the world—and reduce reliance on diesel imports. It leads efforts to diversify electricity generation in favor of clean, renewable sources such as hydro and solar power. It has the potential to cut by more than half of annual greenhouse gas emission. The project will consist of four components: (i) hydropower facility (estimated cost: US$185.6 million); (ii) access road (US$25 million); (iii) transmission line (US$22.8 million); and (iv) technical assistance (US$7.0 million).

The project is supported by many development partners. It will be completed over 2018–22.

Financing Scheme
(In millions of US dollar)

	Total	Loan	Grant
GCF	86.0	70.0	16.0
WB/IDA	33.6	24.7	8.9
ADB	30.0	18.0	12.0
IRENA/ADFD	15.0	15.0	
EDCF	31.6	31.6	
Australia	11.7		11.7
Total	207.9	159.3	48.6

Source: Consideration of funding proposals - Addendum VII, Green Climate Fund.

Disbursement Schedule (TRHDP)
(In millions of US dollar)

	Total	2018	2019	2020	2021
GCF	70.0	28.0	14.0	14.0	14.0
WB/IDA	24.7	8.0	4.0	6.4	6.4
ADB	18.0	7.2	3.6	3.6	3.6
IRENA/ADFD	15.0	1.7	0.9	6.2	6.2
EDCF	31.6	12.6	6.3	6.3	6.3
Total	159.3	57.5	28.8	36.5	36.5

Source: Consideration of funding proposals - Addendum VII, Green Climate Fund.

ASSUMPTIONS UNDER THE BASELINE AND CUSTOMIZED SCENARIOS

A. The Baseline Scenario

6. The macroeconomic assumptions in the DSA reflect developments through mid-2017 and incorporate the risk of natural disasters and climate change over the longer-term. The baseline scenario is based on current policies and is consistent with the macroeconomic framework presented in the Staff Report. The years 2017–22 are assumed to be disaster free to simplify the policy discussion of the near-term outlook. However, from 2023 onwards, the baseline includes the average long-term effects of natural disasters and climate change by lowering GDP growth, raising imports and increasing the fiscal deficit. The discount rate used to calculate the net present value of external debt remains at 5 percent. The main assumptions are:

- **Real GDP growth** is projected at 2.9 percent on average during 2017–22 reflecting the net impact of three factors—(i) a push for higher infrastructure spending; offset by: (ii) continuing public financial management problems which have resulted in domestic arrears and are impeding private sector activity; and (iii) slightly declining logging production. In the long-term, growth is expected to rise to 3.2 percent during 2023–37 due to positive spillovers from infrastructure development, and an increase in mining activity which more than offsets the long-term decline of the forestry sector. This includes the average likely effect of natural disasters and climate change which lowers the growth rate by 0.3 percentage points annually (without natural disasters growth would be around 3.5 percent).
- **Inflation** (measured by GDP deflator in USD terms) is projected to average 3.2 percent during 2017–22, and 4.3 percent during 2023–37 in line with a pickup in real GDP growth to trend and is a little lower than its historical average (of 5.7 percent).
- **Non-interest current account deficit** is projected to widen to 5.6 percent of GDP on average over 2017–22 as infrastructure projects (including the TRHDP and the undersea cable project) have a high import content. The reopening of the Gold Ridge Mine and the resumption of exports is now expected to be delayed until 2023. The overall current account deficit is expected to increase to 6.1 percent of GDP on average during 2023–37, mainly reflecting an assumption that the probability of natural disasters and climate change adds 0.4 percentage points to the deficit annually.
- **FDI inflows** are expected to increase on average to about 3.6 percent of GDP over the 2017–22 based on current infrastructure spending plans and are expected to remain at around 3.4 percent of GDP over the long term, 2023–37.
- **Logging** output is expected to be slightly lower in the next couple of years and then to start declining on average by 1.1 percent a year from 2023. The Gold Ridge Mine is not assumed to reopen quickly (and so is an upside risk to our baseline projection). Over the longer term, gold production is assumed to resume in 2023 and is assumed to peak at 62,000 ounces per annum from 2024 to 2027 and then to decrease gradually. The mine's production is assumed to run for 10 years with a cumulative output of 445,000 ounces, slightly lower than it was assumed in the previous DSA as the continued delay in rehabilitating the mine is likely to have increased the cost of repairs and reduced the economically feasible level of production. Other mining activity (nickel and bauxite) is expected to come fully onstream in the long run, but this is implicitly assumed into long-term growth rates, given the lack of complete information on productive capacity and production projections.
- **External borrowing and grants**: Disbursements for projects in the pipeline supported by the IDA and the ADB are expected to take place in the next five years (2017–21).[2] From 2022 onwards, new external borrowing is projected to average 3.1 percent of GDP, the same level as during 2017–21. Aid flows are expected to grow over the medium term due to the scale-up of IDA and ADB financing but to fall back over the longer run with the grant element of new borrowing declining to about 38

[2] For TRHDP, the disbursement schedule is taken from the Green Climate Fund paper.

percent by 2037. Revenues excluding grants fall to around 29.5 percent of GDP over the long term compared to around 31 percent of GDP in the previous DSA.

- **Fiscal outlook**: The primary deficit is expected to deteriorate from 3.2 percent of GDP in 2016 to 4.0 percent in 2017 and 5.5 percent in 2018 due to an increase in spending on the undersea cable project and TRHDP. The deficit is expected to remain high, averaging 3.7 percent of GDP during 2023–37, due to lower revenues from logging and mining and a high level of development spending. From 2023 onwards, the long-term effect of natural disasters and climate change widens the deficit by 0.2 percentage points annually. The deficit is expected to be financed by both domestic and external borrowing.

Solomon Islands: Baseline Macroeconomic Assumptions
(In percent of GDP, unless otherwise stated)

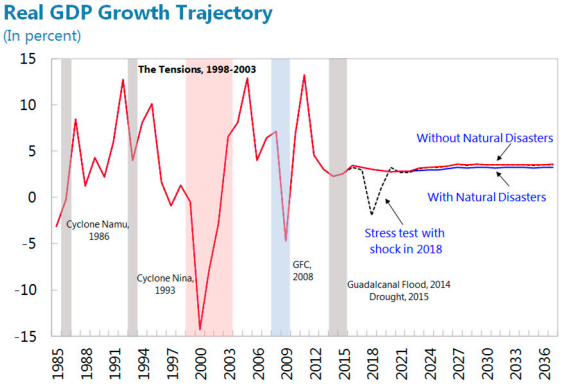
								2017 DSA			2016 DSA	
	2016	2017	2018	2019	2020	2021	2022	2017-22 ave.	2023-37 ave.	Effect of ND and CC	2016-21 ave.	2022-36 ave.
Real GDP growth	3.5	3.2	3.0	2.9	2.8	2.8	2.8	2.9	3.2	0.3	3.1	3.5
GDP deflator in US dollar terms (change in percent)	3.0	0.2	4.6	3.2	3.5	3.8	3.9	3.2	4.3		2.3	4.0
Non-interest current account deficit	3.8	4.3	4.9	6.3	6.6	5.8	6.0	5.6	6.1	0.4	5.4	6.1
Net FDI (negative = inflow)	-2.9	-4.6	-2.5	-3.6	-3.7	-3.8	-3.6	-3.6	-3.4		-3.4	-3.4
Primary deficit	3.2	4.0	5.5	3.6	3.2	2.9	2.8	3.7	3.7	0.2	-0.7	-1.4

Source: IMF staff projections.

B. The Customized Scenario with A One-off Natural Disaster Shock

7. A one-off severe natural disaster shock is assumed to occur in 2018 adding to external borrowing as well as adjusting GDP growth and the current account. Solomon Islands, like other Pacific island countries, is highly vulnerable to natural disasters. EM-DAT, the international disaster database, shows that for the past twenty-seven years (1980-2016), the country's largest damage from natural disasters was 14 percent of GDP. The customized scenario assumes a natural disaster of this scale hits in 2018 and the same amount of funds for recovery and reconstruction is financed by a mix of grants (4.0 percent of GDP) and concessional loans (10.0 percent of GDP). GDP growth falls by around 5.0 percentage points in 2018—a similar swing to that seen in previous large disaster years (excluding events during the tensions and the global financial crisis) and is expected to rise by 3.0 percentage points in 2019 and

Real GDP Growth Trajectory
(In percent)



Sources: IMF staff estimates.

Solomon Islands: Macroeconomic Assumptions Under the Customized Scenario
(In percent of GDP, unless otherwise stated)

Deviation from the baseline	2018	2019	2020
GDP growth (percentage points)	-5.0%	3.0%	2.2%
Current account balance	-3.0%	-4.0%	-4.0%
Financing	14.0%		
Grants	4.0%		
Concessional Loans	10.0%		

Source: IMF staff projections.

2.2 percentage points in 2020, with the level of GDP returning to the baseline in 2020. The current account deficit is assumed to be higher over 2018–20 due to damage to the export base and increases in imports for reconstruction. From 2023 onwards the shock scenario includes the average long-term effect of natural disasters.

8. The scenario highlights that while the debt level is currently low, Solomon Islands is highly susceptible to shocks and a sizeable buffer is needed to cope with unexpected events. The recent deterioration of the fiscal position and rundown of the cash balance leave the authorities ill-equipped to handle even a small shock.

DEBT SUSTAINABILITY

A. External Debt Sustainability Analysis

9. Under the baseline scenario, all PV of external PPG debt indicators remain below the policy relevant thresholds (Figure 1). Total external PPG debt is projected to gradually increase from 7.5 percent of GDP in 2016 to 18.2 percent in 2021 due mainly to new disbursements from the TRHDP project and continue to grow to 30.1 percent of GDP in 2037 (Table 1).

10. However, the sensitivity analysis shows that the debt dynamics deteriorate under alternative scenarios with less favorable terms on new borrowing having the largest impact. A shock to financing terms— a rise in interest rates on new borrowing by 2 percentage points compared to the baseline— remains the most important risk to sustainability, causing the breach of the indicative threshold of the PV of debt-to-GDP ratio and the PV of debt-to-export ratio (Figure 1). This suggests that external borrowing should be contracted on concessional terms as much as possible to contain the debt burden.

11. The severe natural disaster shock causes all the debt trajectories for each external debt indicator to move upward, indicating that debt sustainability has deteriorated after the shock. Given the current low level of external debt, the indicators remain below thresholds. However, while two severe shocks are unlikely to hit within a twenty-year horizon, the country could be hit by two bigger than average shocks—as the probability of a disaster causing more than 3 percent of GDP in damage and affecting more than 5 percent of the population is likely to occurs every seven years—and this could have a larger cumulative effect on debt sustainability, especially if it results in a prolonged period of low growth.

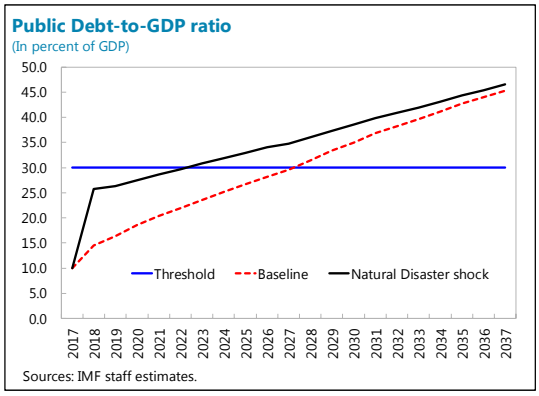
B. Public Sector Debt Sustainability Analysis

12. The PV of public sector debt in percent of GDP remains below the threshold under the baseline scenario (Figure 2). However, the public debt-to-GDP ratio continues to rise from 7.9 percent and on current policies breaches the authorities' threshold of 30 percent in 2028 (Table 4). The breach is primarily driven by a high disbursement schedule and continued expansionary fiscal policy.

13. Sensitivity analysis also reveals that public debt sustainability remains vulnerable to shocks, particularly to the severe shock to real GDP growth (Figure 2). The PV of debt-to-GDP ratio breaches the 38 percent of GDP benchmark by 2026, highlighting the need for stronger growth in the long run.

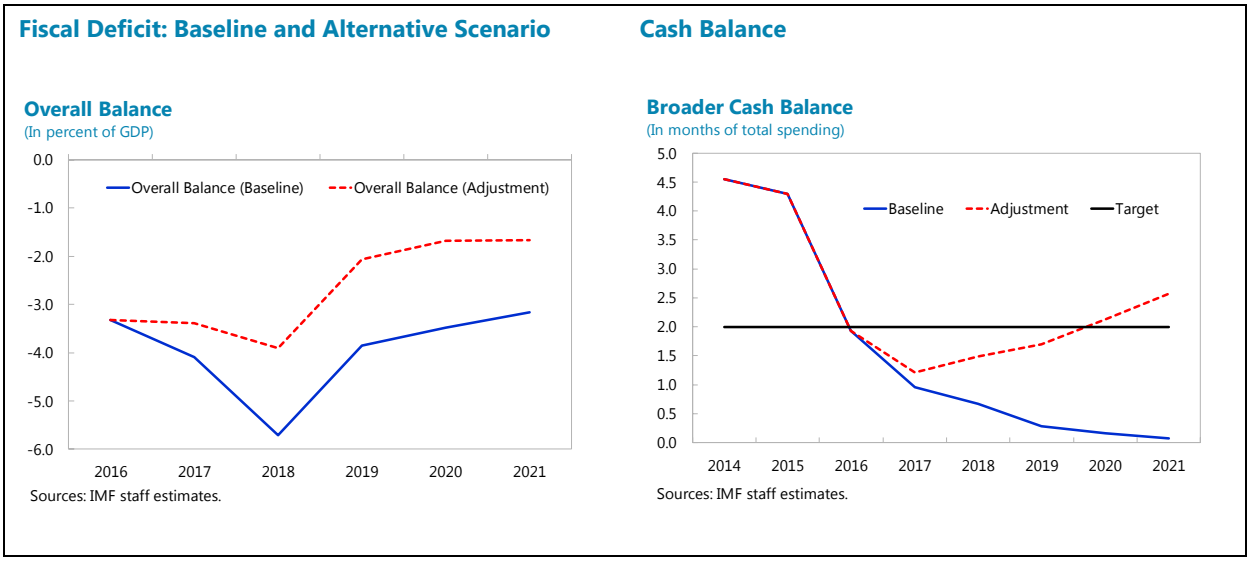
14. The severe natural disaster shock results in a sharper deterioration in debt sustainability. The public debt-to-GDP ratio would breach the authorities' threshold of 30 percent in 2023, five years earlier than in the baseline scenario.

15. The breach of the public debt threshold set by the authorities under the baseline and natural disaster scenarios highlights the importance of fiscal consolidation over the medium term. As shown in Figure 2 in the Annex for the fixed primary balance scenario, which assumes an unchanged primary balance from 2016 for the entire projection period, the PV of the debt-to-GDP ratio, the PV of the debt-to-revenue ratio, and debt service-to-revenue ratio would be higher than the baseline over the projection period.



C. Policy Adjustment Scenario

16. As highlighted in the staff report, fiscal policy needs to be placed on a firmer footing, first by rebuilding the cash balance. The policy adjustment scenario demonstrates how this can be achieved. While debt levels are similar in the early years, as cash buffers are rebuilt, debt levels are lower over the longer-term (see Figure 3 which illustrated the policy adjustment scenario).



CONCLUSION

17. The current DSA suggests that Solomon Islands' risk of external debt distress continues to be moderate. The baseline scenario takes into account the long-term effects of natural disasters and climate change by adjusting real GDP growth, the current account balance, and fiscal balance from 2023 onwards. While there is no breach of thresholds under the baseline, the stress test with a shock to financing terms on new borrowing indicates that the PV of external debt-to-GDP ratio and the PV of external

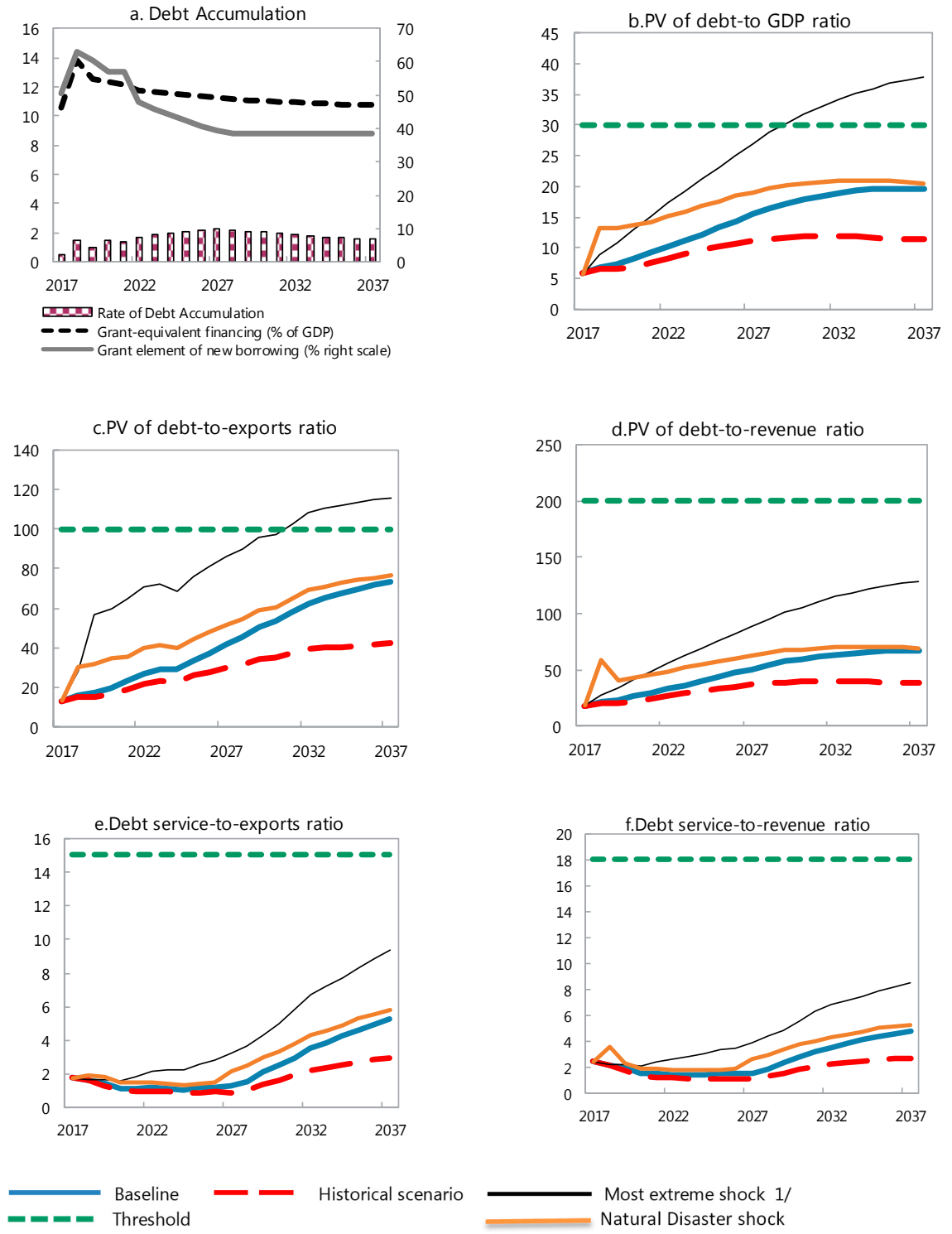
debt-to-exports ratio breach the respective policy-dependent indicative debt thresholds for some years. Overall the analysis illustrates the vulnerability of debt sustainability to external financing, fiscal and growth shocks and the important contribution of aid flows and FDI to supporting debt sustainability. This highlights the importance of being cautious on expenditures, prioritizing investment projects that build resilience to natural disasters and other investments that can help boost potential growth and broaden the export base or reduce reliance on imports. To keep the debt burden contained, external borrowing should be contracted on concessional terms as much as possible.

18. Public debt is on a rising trajectory and under current policies could breach the threshold set by the authorities at 30 percent of GDP in 2028, indicating the need to tighten fiscal policy, strengthen public financial management, and boost the long-term economic growth. A shock to real GDP growth has the largest impact on public debt sustainability, leading the PV of debt-to-GDP ratio to breach the 38 percent of GDP benchmark by 2030. The customized scenario, with a one-off severe natural disaster shock, also shows that debt sustainability can quickly deteriorate in response to a shock. Given the vulnerability to natural disaster shocks, the authorities need to recalibrate fiscal policy first to rebuild a fiscal buffer to enable smoother month-to-month cash management and this is likely to require both stronger revenue mobilization efforts and expenditure rationalization.

AUTHORITIES' VIEW

19. The authorities broadly agree with the assessment of debt sustainability analysis and welcome the incorporation of the average long-term effects of natural disasters and climate change into the baseline. In the medium term, they intend to increase both external borrowing, mainly for the Tina River hydropower project, and domestic borrowing for large infrastructure projects, especially for the undersea cable project. The authorities noted that they are mindful of the burden of external debt, and given their vulnerability to changes in financing terms, they would like to seek external concessional and preferably grant finance for the bulk of external assistance. The authorities noted the uncertainty over the quality of nominal GDP data and the different estimates of the Ministry of Finance and IMF staff. They noted the IMF staff's response that the differences in nominal GDP estimates were not large enough to impact on the risk rating. They highlighted that the National Statistics Office is working on revising and strengthening the GDP data and that the new data will be published early next year.

Figure 1. Solomon Islands: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2017–37 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2027. In figure b. it corresponds to a Terms shock; in c. to a Exports shock; in d. to a Terms shock; in e. to a Terms shock and in figure f. to a Terms shock

Table 1. Solomon Islands: External Debt Sustainability Framework, Baseline Scenario, 2014–37 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections													
	2014	2015	2016			2017	2018	2019	2020	2021	2022	2017-2022 Average	2023	2024	2025	2026	2027	2037	2023-2037 Average
External debt (nominal) 1/	**12.0**	**10.6**	**8.2**			**9.0**	**12.8**	**14.9**	**17.3**	**19.3**	**21.1**		**22.6**	**24.1**	**25.5**	**26.9**	**28.0**	**32.1**	
of which: public and publicly guaranteed (PPG)	9.9	9.7	7.5			8.2	11.9	13.9	16.2	18.2	19.9		21.3	22.8	24.1	25.4	26.5	30.3	
Change in external debt	-7.2	-1.4	-2.4			0.8	3.8	2.1	2.4	2.0	1.8		1.5	1.5	1.4	1.4	1.1	-0.1	
Identified net debt-creating flows	1.8	0.8	0.4			-0.4	2.3	2.4	2.6	1.7	2.1		1.9	2.0	2.2	2.4	2.4	1.9	
Non-interest current account deficit	**3.8**	**2.8**	**3.8**	10.4	10.7	**4.3**	**4.9**	**6.3**	**6.6**	**5.8**	**6.0**	5.6	**5.7**	**5.9**	**6.0**	**6.2**	**6.4**	**5.8**	6.1
Deficit in balance of goods and services	9.9	8.1	5.8			6.6	8.3	10.1	10.5	11.7	12.4		12.2	11.3	11.9	12.5	13.1	12.5	
Exports	48.3	45.5	44.9			44.2	42.8	42.4	41.7	39.8	38.1		38.5	42.2	39.5	38.5	37.2	26.8	
Imports	58.2	53.5	50.7			50.8	51.2	52.5	52.2	51.5	50.4		50.6	53.4	51.5	51.0	50.3	39.4	
Net current transfers (negative = inflow)	-8.7	-7.1	-5.2	-14.4	6.7	-5.9	-6.7	-7.0	-7.2	-7.7	-8.0	-7.1	-8.5	-8.7	-8.8	-9.0	-9.1	-6.9	-8.9
of which: official	-8.5	-8.1	-7.1			-7.5	-8.2	-8.0	-7.9	-8.1	-8.1		-8.3	-8.2	-8.2	-8.2	-8.1	-5.1	
Other current account flows (negative = net inflow)	2.5	1.9	3.2			3.5	3.3	3.1	3.3	1.8	1.6		2.0	3.3	3.0	2.7	2.5	0.1	
Net FDI (negative = inflow)	**-1.7**	**-2.4**	**-2.9**	-8.4	7.3	**-4.6**	**-2.5**	**-3.6**	**-3.7**	**-3.8**	**-3.6**	-3.6	**-3.5**	**-3.5**	**-3.5**	**-3.4**	**-3.5**	**-3.4**	-3.4
Endogenous debt dynamics 2/	**-0.2**	**0.4**	**-0.5**			**-0.1**	**-0.1**	**-0.2**	**-0.2**	**-0.3**	**-0.3**		**-0.3**	**-0.4**	**-0.4**	**-0.4**	**-0.5**	**-0.5**	
Contribution from nominal interest rate	0.5	0.2	0.1			0.1	0.1	0.2	0.2	0.2	0.2		0.2	0.3	0.3	0.3	0.3	0.5	
Contribution from real GDP growth	-0.4	-0.3	-0.3			-0.3	-0.3	-0.4	-0.4	-0.5	-0.5		-0.6	-0.6	-0.7	-0.7	-0.8	-1.0	
Contribution from price and exchange rate changes	-0.3	0.5	-0.3			…	…	…	…	…	…		…	…	…	…	…	…	
Residual (3-4) 3/	**-9.0**	**-2.3**	**-2.7**			**1.2**	**1.5**	**-0.4**	**-0.2**	**0.3**	**-0.3**		**-0.4**	**-0.5**	**-0.8**	**-1.0**	**-1.3**	**-2.0**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	
PV of external debt 4/	…	…	6.6			6.7	7.7	8.4	9.4	10.3	11.4		12.5	13.6	14.7	15.9	17.0	21.6	
In percent of exports	…	…	14.6			15.1	18.1	19.8	22.5	25.9	29.9		32.4	32.3	37.3	41.3	45.6	80.3	
PV of PPG external debt	**…**	**…**	**5.8**			**5.8**	**6.8**	**7.4**	**8.3**	**9.2**	**10.2**		**11.2**	**12.2**	**13.3**	**14.4**	**15.5**	**19.7**	
In percent of exports	**…**	**…**	**12.9**			**13.1**	**15.9**	**17.5**	**20.0**	**23.0**	**26.7**		**29.1**	**29.1**	**33.7**	**37.4**	**41.5**	**73.3**	
In percent of government revenues	**…**	**…**	**18.3**			**18.3**	**21.6**	**23.6**	**26.7**	**29.5**	**32.9**		**36.5**	**39.9**	**43.4**	**47.2**	**50.8**	**66.8**	
Debt service-to-exports ratio (in percent)	**16.5**	**4.9**	**3.6**			**2.0**	**1.8**	**1.6**	**1.3**	**1.3**	**1.3**		**1.3**	**1.2**	**1.3**	**1.4**	**1.4**	**5.6**	
PPG debt service-to-exports ratio (in percent)	**1.6**	**1.5**	**1.8**			**1.8**	**1.6**	**1.4**	**1.1**	**1.2**	**1.2**		**1.2**	**1.1**	**1.2**	**1.2**	**1.3**	**5.3**	
PPG debt service-to-revenue ratio (in percent)	**2.3**	**2.0**	**2.6**			**2.5**	**2.2**	**1.9**	**1.5**	**1.5**	**1.5**		**1.5**	**1.5**	**1.5**	**1.6**	**1.5**	**4.8**	
Total gross financing need (Millions of U.S. dollars)	117.4	31.0	30.9			6.8	43.3	48.6	52.9	40.7	51.8		52.0	59.7	67.8	78.8	87.4	203.9	
Non-interest current account deficit that stabilizes debt ratio	10.9	4.3	6.2			3.5	1.1	4.2	4.2	3.7	4.2		4.1	4.3	4.6	4.8	5.2	5.8	
Key macroeconomic assumptions																			
Real GDP growth (in percent)	2.3	2.5	3.5	4.5	4.6	3.2	3.0	2.9	2.8	2.8	2.8	2.9	2.9	3.0	3.0	3.1	3.3	3.3	3.2
GDP deflator in US dollar terms (change in percent)	1.6	-3.8	3.0	5.7	6.7	0.2	4.6	3.2	3.5	3.8	3.9	3.2	4.5	4.1	4.3	4.0	4.5	4.5	4.3
Effective interest rate (percent) 5/	2.7	1.6	1.3	2.4	0.6	1.7	1.8	1.4	1.3	1.2	1.2	1.4	1.2	1.3	1.3	1.3	1.4	1.7	1.5
Growth of exports of G&S (US dollar terms, in percent)	-1.1	-7.1	5.4	15.2	25.8	1.7	4.4	5.2	4.7	1.7	2.2	3.3	8.6	17.4	0.7	4.4	4.4	5.6	5.1
Growth of imports of G&S (US dollar terms, in percent)	-4.0	-9.3	1.0	10.8	21.8	3.6	8.6	9.0	5.7	5.2	4.7	6.1	7.9	13.1	3.5	6.3	6.4	6.8	5.8
Grant element of new public sector borrowing (in percent)	…	…	…	…	…	50.7	62.8	60.5	56.9	56.9	48.0	56.0	46.0	44.1	42.3	40.8	39.5	38.4	39.8
Government revenues (excluding grants, in percent of GDP)	32.8	35.1	31.7			31.7	31.5	31.4	31.2	31.0	30.9		30.7	30.7	30.7	30.6	30.4	29.5	
Aid flows (in Millions of US dollars) 7/	170.4	149.0	117.2			147.3	212.8	200.5	209.3	218.6	219.5		232.2	245.0	259.5	274.6	292.6	582.5	
of which: Grants	170.4	149.0	117.2			121.7	148.9	154.4	160.9	170.7	181.5		194.1	207.0	221.2	236.0	253.3	504.7	
of which: Concessional loans	0.0	0.0	0.0			25.7	63.9	46.1	48.4	47.9	38.1		38.0	38.0	38.3	38.6	39.3	77.8	
Grant-equivalent financing (in percent of GDP) 8/	…	…	…			10.5	13.8	12.5	12.3	12.1	11.7	12.2	11.6	11.5	11.4	11.3	11.2	10.7	11.1
Grant-equivalent financing (in percent of external financing) 8/	…	…	…			91.4	88.6	90.8	89.1	89.6	87.8	89.6	87.3	86.8	86.3	85.9	85.5	84.8	85.4
Memorandum items:																			
Nominal GDP (Millions of US dollars)	1174.0	1158.4	1235.0			1277.0	1376.8	1462.8	1555.6	1659.3	1772.5		1905.9	2041.9	2193.4	2351.8	2537.1	5288.6	
Nominal dollar GDP growth	3.9	-1.3	6.6			3.4	7.8	6.2	6.3	6.7	6.8	6.2	7.5	7.1	7.4	7.2	7.9	7.9	7.6
PV of PPG external debt (in Millions of US dollars)			69.2			75.4	93.9	108.3	129.8	151.9	180.4		213.0	250.1	292.0	339.1	392.2	1040.4	
(PVt-PVt-1)/GDPt-1 (in percent)						0.5	1.4	1.0	1.5	1.4	1.7	1.3	1.8	1.9	2.1	2.1	2.3	1.5	1.9
Gross workers' remittances (Millions of US dollars)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	
PV of PPG external debt (in percent of GDP + remittances)	…	…	5.8			5.8	6.8	7.4	8.3	9.2	10.2		11.2	12.2	13.3	14.4	15.5	19.7	
PV of PPG external debt (in percent of exports + remittances)	…	…	12.9			13.1	15.9	17.5	20.0	23.0	26.7		29.1	29.1	33.7	37.4	41.5	73.3	
Debt service of PPG external debt (in percent of exports + remittances)	…	…	1.8			1.8	1.6	1.4	1.1	1.2	1.2		1.2	1.1	1.2	1.2	1.3	5.3	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes. The residuals in 2014 was due largely to net FDI, which was revised down as a result of a changes to ensure the correct treatment of net losses under reinvested earnings, in line with BPM6.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Solomon Islands: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt 2017–37

(In percent)

	Projections							
	2017	2018	2019	2020	2021	2022	**2027**	2037
PV of debt-to GDP ratio								
Baseline	6	7	7	8	9	10	**15**	20
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	6	7	7	7	8	8	11	11
A2. New public sector loans on less favorable terms in 2017-2037 2	6	9	11	13	15	17	**27**	38
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	6	7	8	9	10	11	**16**	21
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	6	10	17	18	19	19	**23**	22
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	6	7	8	9	10	11	**17**	22
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	6	7	9	10	10	11	**17**	20
B5. Combination of B1-B4 using one-half standard deviation shocks	6	4	3	4	5	7	**13**	19
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	6	10	10	12	13	14	**22**	28
PV of debt-to-exports ratio								
Baseline	13	16	17	20	23	27	**42**	73
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	13	15	15	16	19	22	30	43
A2. New public sector loans on less favorable terms in 2017-2037 2	13	21	25	31	38	45	**72**	141
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	13	16	17	20	23	27	**42**	73
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	13	28	57	60	65	70	**86**	115
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	13	16	17	20	23	27	**42**	73
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	13	17	21	23	26	30	**44**	75
B5. Combination of B1-B4 using one-half standard deviation shocks	13	9	8	11	14	17	**35**	73
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	13	16	17	20	23	27	**42**	73
PV of debt-to-revenue ratio								
Baseline	18	22	24	27	30	33	**51**	67
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	18	21	21	22	24	27	37	39
A2. New public sector loans on less favorable terms in 2017-2037 2	18	28	34	42	49	56	**89**	129
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	18	22	25	28	31	35	**54**	71
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	18	33	55	58	60	63	**76**	76
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	18	23	26	29	33	36	**56**	74
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	18	23	28	31	34	37	**54**	68
B5. Combination of B1-B4 using one-half standard deviation shocks	18	13	11	14	18	21	**42**	66
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	18	30	33	37	41	46	**71**	94

Table 3. Solomon Islands: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2017–37 (continued)

(in percent)

Debt service-to-exports ratio								
Baseline	2	2	1	1	1	1	**1**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	2	2	1	1	1	1	**1**	3
A2. New public sector loans on less favorable terms in 2017-2037 2	2	2	2	2	2	2	**3**	9
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	2	2	1	1	1	1	**1**	5
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	2	2	2	2	2	2	**3**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	2	2	1	1	1	1	**1**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	2	2	1	1	1	1	**1**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	2	2	1	1	1	1	**1**	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	2	2	1	1	1	1	**1**	5
Debt service-to-revenue ratio								
Baseline	2	2	2	2	2	1	**2**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	2	2	2	1	1	1	**1**	3
A2. New public sector loans on less favorable terms in 2017-2037 2	2	2	2	2	2	3	**4**	9
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	2	2	2	2	2	2	**2**	5
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	2	2	2	2	2	2	**3**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	2	2	2	2	2	2	**2**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	2	2	2	2	2	2	**2**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	2	2	2	1	1	1	**1**	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	2	3	3	2	2	2	**2**	7
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	43	43	43	43	43	43	**43**	43

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

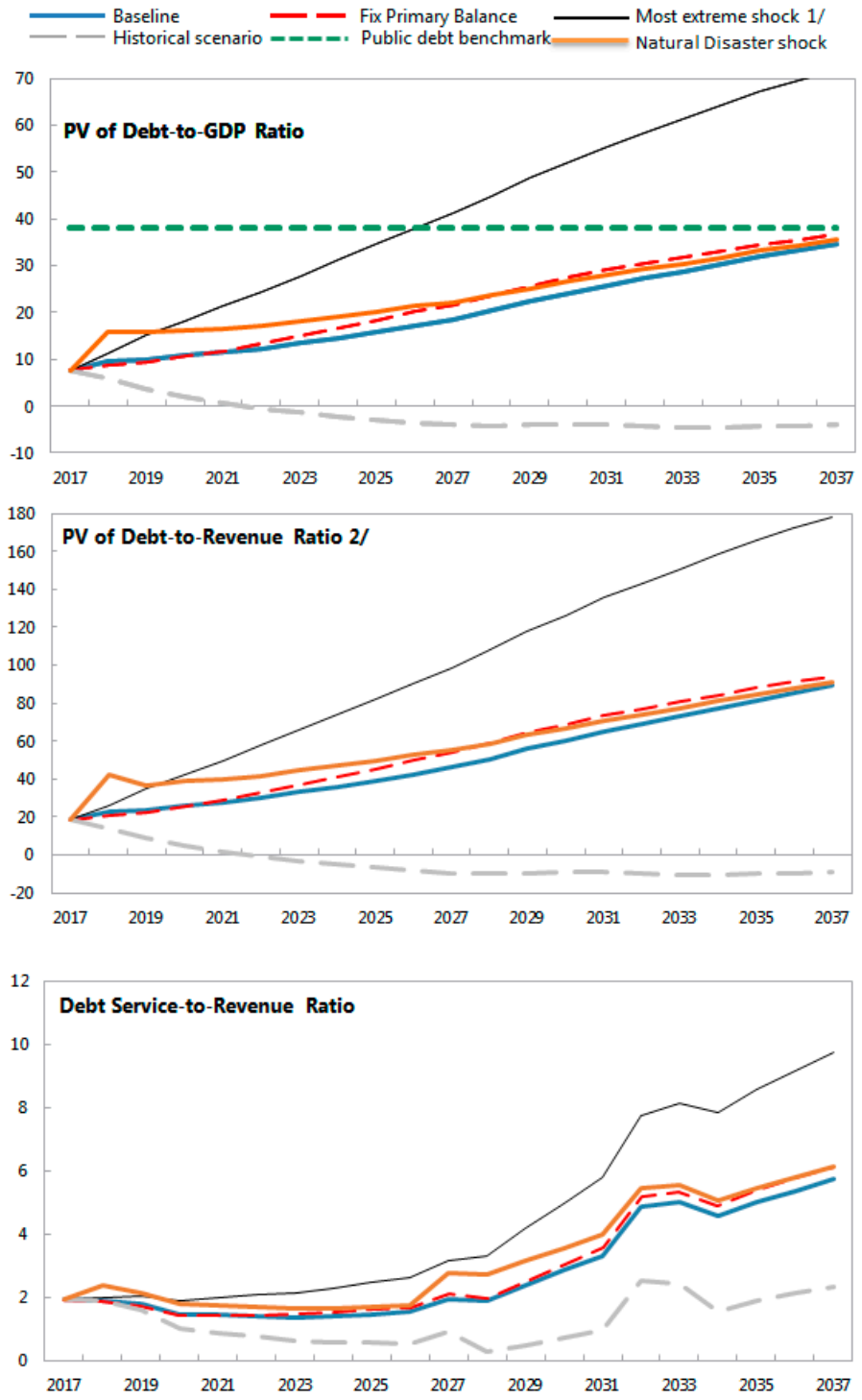
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 2. Solomon Islands; Indicators of Public Debt Under Alternative Scenarios, 2017–37 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2027.
2/ Revenues are defined inclusive of grants.

Table 4. Solomon Islands; Public Sector Debt Sustainability Framework, baseline Scenario, 2014–37

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2014	2015	2016	Average 5/	Standard Deviation 5/	2017	2018	2019	2020	2021	2022	2017-22 Average	2027	2037	2023-37 Average
Public sector debt 1/	12.8	10.1	7.9			10.0	14.6	16.4	18.6	20.4	21.9		29.6	45.3	
of which: foreign-currency denominated	9.9	9.7	7.5			8.2	11.9	13.9	16.2	18.2	19.9		26.5	30.3	
Change in public sector debt	-2.5	-2.6	-2.2			2.1	4.5	1.9	2.2	1.8	1.5		1.4	1.3	
Identified debt-creating flows	-2.4	0.4	2.8			3.5	5.2	3.0	2.5	2.0	1.7		1.7	2.3	
Primary deficit	-1.8	0.1	3.2	-2.8	3.4	4.0	5.5	3.6	3.2	2.9	2.8	3.7	3.3	4.3	3.7
Revenue and grants	47.3	47.9	41.2			41.2	42.4	42.0	41.6	41.3	41.1		40.4	39.0	
of which: grants	14.5	12.9	9.5			9.5	10.8	10.6	10.3	10.3	10.2		10.0	9.5	
Primary (noninterest) expenditure	45.5	48.0	44.4			45.2	47.9	45.6	44.8	44.2	43.9		43.7	43.3	
Automatic debt dynamics	-0.5	0.3	-0.4			-0.5	-0.4	-0.6	-0.7	-0.9	-1.0		-1.6	-2.0	
Contribution from interest rate/growth differential	-0.4	-0.4	-0.4			-0.3	-0.3	-0.4	-0.5	-0.6	-0.6		-0.9	-1.2	
of which: contribution from average real interest rate	-0.1	-0.1	-0.1			0.0	0.0	0.0	0.0	-0.1	-0.1		-0.1	0.2	
of which: contribution from real GDP growth	-0.3	-0.3	-0.3			-0.2	-0.3	-0.4	-0.4	-0.5	-0.6		-0.9	-1.4	
Contribution from real exchange rate depreciation	-0.1	0.8	0.0			-0.3	-0.1	-0.2	-0.2	-0.3	-0.4		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes 6/	-0.1	-3.1	-5.0			-1.4	-0.6	-1.1	-0.3	-0.2	-0.2		-0.3	-1.0	
Other Sustainability Indicators															
PV of public sector debt	...	...	6.2			7.6	9.5	9.9	10.7	11.4	12.2		18.6	34.7	
of which: foreign-currency denominated	...	...	5.8			5.8	6.8	7.4	8.3	9.2	10.2		15.5	19.7	
of which: external	...	...	5.8			5.8	6.8	7.4	8.3	9.2	10.2		15.5	19.7	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	-0.2	2.7	4.5			5.2	6.7	4.7	4.2	3.8	3.6		4.3	6.6	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	15.1			18.6	22.4	23.6	25.7	27.5	29.8		45.9	89.1	
PV of public sector debt-to-revenue ratio (in percent)	...	...	19.6			24.2	30.1	31.5	34.3	36.7	39.6		61.0	117.9	
of which: external 3/	...	...	18.3			18.3	21.6	23.6	26.7	29.5	32.9		50.8	66.8	
Debt service-to-revenue and grants ratio (in percent) 4/	2.5	4.5	2.0			1.9	1.9	1.8	1.5	1.4	1.4		1.9	5.7	
Debt service-to-revenue ratio (in percent) 4/	3.5	6.2	2.6			2.5	2.5	2.4	2.0	1.9	1.9		2.6	7.6	
Primary deficit that stabilizes the debt-to-GDP ratio	0.7	2.7	5.5			1.9	1.0	1.7	1.0	1.1	1.2		2.0	3.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	2.3	2.5	3.5	4.5	4.6	3.2	3.0	2.9	2.8	2.8	2.8	2.9	3.3	3.3	3.2
Average nominal interest rate on forex debt (in percent)	1.1	0.9	1.0	1.3	0.6	1.4	1.4	1.1	1.0	0.9	0.9	1.1	1.2	1.5	1.3
Average real interest rate on domestic debt (in percent)	-1.4	-1.8	...	-4.2	3.6	-0.7	2.5	2.4	2.2	1.9	1.8	1.7	1.5	1.9	1.9
Real exchange rate depreciation (in percent, + indicates depreciation)	-1.0	8.1	-0.4	-3.2	6.9	-3.6	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	2.5	3.1	3.6	6.0	4.1	1.6	2.7	3.2	3.5	3.8	3.9	3.1	4.5	4.5	4.3
Growth of real primary spending (deflated by GDP deflator, in percent)	-0.3	8.3	-4.3	0.4	3.1	5.0	9.2	-2.0	1.0	1.4	2.1	2.8	3.1	3.2	3.1
Grant element of new external borrowing (in percent)	...	...	...	...	...	50.7	62.8	60.5	56.9	56.9	48.0	56.0	39.5	38.4	...

Sources: Country authorities; and staff estimates and projections.

1/ Coverage of public sector debt is general government and gross debt.

Change in assets in 2015 and 2016 reflects the repayment of public domestic debt

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

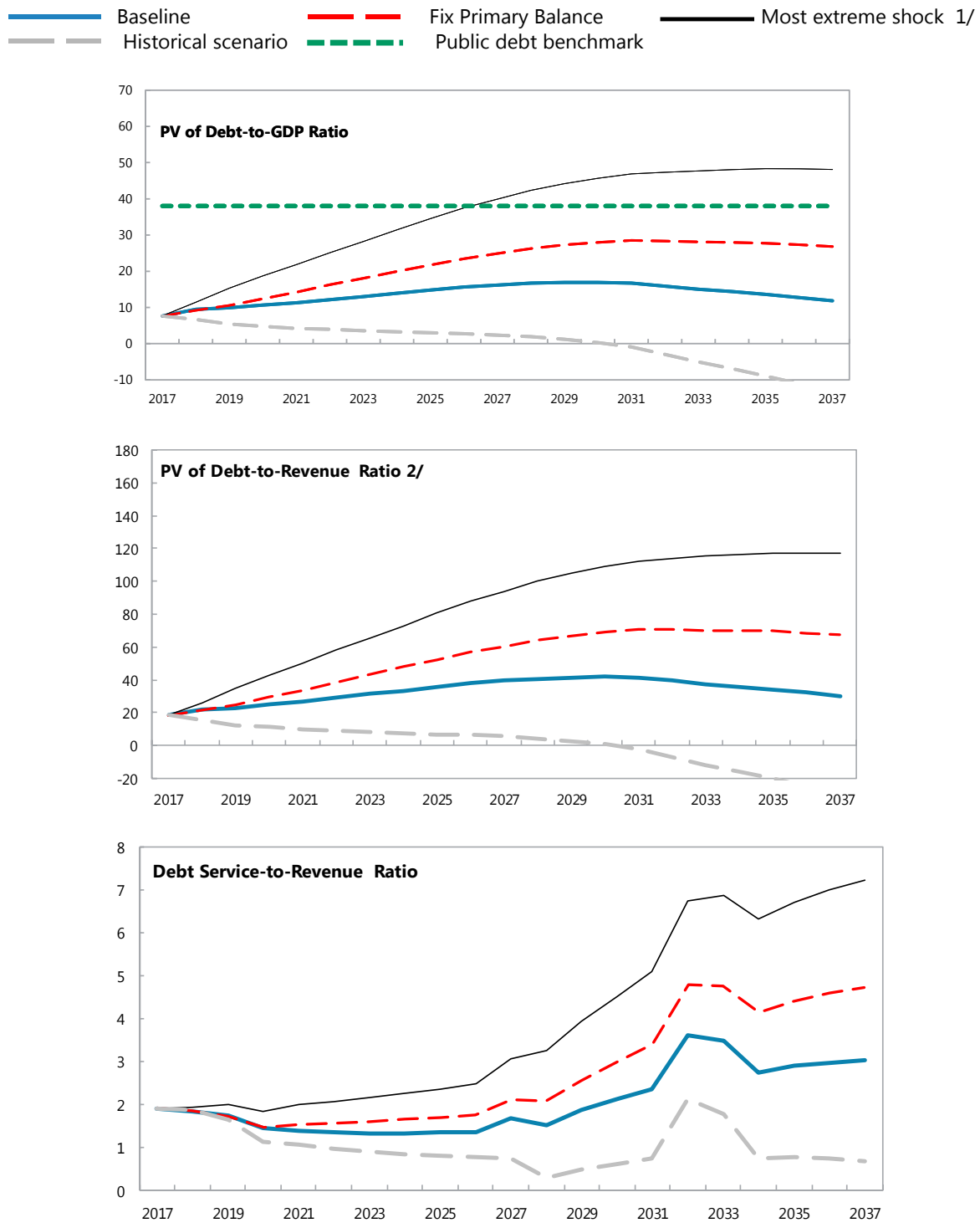
6/ The residuals in 2016 was due largely to a decline in cash reserves of 2.9 percent of GDP.

Table 5. Solomon Islands: Sensitivity Analysis is for Key Indicators of Public Debt 2017–37

	Projections							
	2017	2018	2019	2020	2021	2022	2027	2037
PV of Debt-to-GDP Ratio								
Baseline	8	9	10	11	11	12	19	35
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	8	6	4	2	1	-1	-4	-4
A2. Primary balance is unchanged from 2017	8	9	9	11	12	13	22	37
A3. Permanently lower GDP growth 1/	8	10	11	12	14	15	29	73
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	8	11	15	18	21	24	41	72
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	8	8	7	8	9	10	16	34
B3. Combination of B1-B2 using one half standard deviation shocks	8	7	6	8	11	13	27	55
B4. One-time 30 percent real depreciation in 2018	8	11	10	10	10	11	16	31
B5. 10 percent of GDP increase in other debt-creating flows in 2018	8	15	15	16	17	17	23	37
PV of Debt-to-Revenue Ratio 2/								
Baseline	19	22	24	26	28	30	46	89
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	19	14	9	5	1	-1	-10	-9
A2. Primary balance is unchanged from 2017	19	20	22	25	29	33	54	94
A3. Permanently lower GDP growth 1/	19	23	25	29	32	37	71	177
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	19	26	35	42	50	57	98	178
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	19	18	17	19	21	24	41	86
B3. Combination of B1-B2 using one half standard deviation shocks	19	16	13	19	25	31	65	138
B4. One-time 30 percent real depreciation in 2018	19	25	24	24	25	26	39	80
B5. 10 percent of GDP increase in other debt-creating flows in 2018	19	36	37	38	40	42	56	95
Debt Service-to-Revenue Ratio 2/								
Baseline	2	2	2	1	1	1	2	6
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	2	2	2	1	1	1	1	2
A2. Primary balance is unchanged from 2017	2	2	2	1	1	1	2	6
A3. Permanently lower GDP growth 1/	2	2	2	2	2	2	3	9
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	2	2	2	2	2	2	3	10
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	2	2	2	1	1	1	2	5
B3. Combination of B1-B2 using one half standard deviation shocks	2	2	2	1	1	1	2	8
B4. One-time 30 percent real depreciation in 2018	2	2	2	2	2	2	2	8
B5. 10 percent of GDP increase in other debt-creating flows in 2018	2	2	2	2	2	2	2	6

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Figure 3. Policy Adjustment Scenario: Indicators of Public Debt Under Alternative Scenarios, 2017–37 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2027.
2/ Revenues are defined inclusive of grants.